

and around 0.50% per quarter. Interestingly, the factor benchmark consisting of just bonds and stocks indicates that the optimal combination of stocks and bonds to mimic real estate is 35% stocks and 65% bonds.

Figure 10.4 graphs cumulative excess returns of direct real estate relative to these factor benchmarks. While there was some value added in the early 1980s relative to these REIT, bond, and stock factors, the factor benchmarks did much better than direct real estate from the mid-1980s to the early 2000s. Direct real estate picked up relative to the factor benchmarks in the mid-2000s, coinciding with the period's property boom. Figure 10.4 clearly shows the crash in real estate markets in 2008 and 2009 toward the end of the sample.

3.3 TIME-VARYING FACTOR EXPOSURES

William Sharpe, one of the inventors of the CAPM, introduced a powerful framework to handle time-varying benchmarks in 1992. He called it “*style analysis*.” In our context, style analysis is a factor benchmark where the factor exposures evolve through time.¹¹

To illustrate time-varying factor exposures in the spirit of Sharpe's style analysis, consider four funds:

LSVEX: LSV Value Equity. LSV is a “quantitative value equity manager providing active management for institutional investors” and was named after its founding academics: Josef Lakonishok, Andrei Shleifer, and Robert Vishny.¹²

FMAGX: Fidelity Magellan. One of the most famous retail mutual funds, it grew to prominence under superstar manager Peter Lynch in the 1980s and 1990s.

GSCGX: Goldman Sachs Capital Growth. How can we not include a Goldman Sachs name?

BRK: Berkshire Hathaway. Since we've been using Buffett's example, let's stay with it.

I use monthly data from January 2001 to December 2011.

Here are the Fama–French and momentum factor regressions using *constant* factor weights:

¹¹ Computing standard errors for alphas when factor loadings vary over time, and even when the alphas themselves vary over time, is tricky, as Ang and Kristensen (2012) show. For a summary of style analysis, see ter Horst, Nijman, and de Roon (2004).

¹² See <http://www.lsvasset.com/about/about.html>.